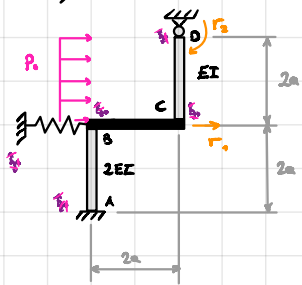


Problem 1)



$$k = \frac{5EI}{8a^3}$$

a) I would say the system has 2 dof, with r_1 being the horizontal displacement of BC.

r_2 is the rotation of the beam CD at point C.

Reasoning: A: Boundary conditions: x_1, z_0, θ_1

B: no axial deformation of AB $\Rightarrow z_0$

CD $\Rightarrow \theta_0$

$$\Rightarrow x_1 = r_1$$

C: no axial deformation of CD $\Rightarrow z_c$

AB $\Rightarrow \theta_c$

$$\Rightarrow x_c = x_0 = r_1$$

D: Boundary conditions: x_0, z_0

$$\Rightarrow \theta_0 = r_2$$

However, as $M_D = 0$, it does not contribute to the stiffness of the system and can be neglected; the element CD can be modeled using a 3dof element $\rightarrow \vec{r} = r_1$ only.

b) Loads that are applied directly on the nodes can be put directly into $\vec{R}^{(n)}$ at the correct dof.

If distributed loads are present, the consistent load vectors $\vec{S}^{(e)}$ need to be calculated.

They can be determined via the reaction forces from the Beam-table. For this, the element is imagined to be clamped at both sides:

$$\left. \begin{array}{l} H_A, H_B = \frac{PL}{2} \\ F_A, F_B = \frac{PL}{2} \end{array} \right\} \vec{S}^{(e)} = \frac{PL}{2} \begin{bmatrix} -1 \\ \frac{L}{6} \\ -1 \\ -\frac{L}{6} \end{bmatrix} = \frac{PL}{12} \begin{bmatrix} -6 \\ L \\ -6 \\ -L \end{bmatrix}$$

With $\vec{S}^{(e)}$ known for every element, the system nodal force vector can be computed via $\vec{R}^{(n)} = \sum_{e=1}^N \vec{a}^{(e)T} \vec{S}^{(e)}$

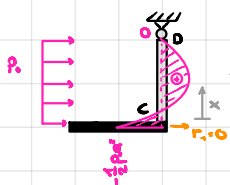
The final load vector $\vec{R}$ is then the difference between the assembled load vector $\vec{R}^{(n)}$ and $\vec{R}^{(e)}$.

$$\vec{R} = \vec{R}^{(n)} - \vec{R}^{(e)}$$

c) Bernoulli beam theory:

- no shear, cross-section always perpendicular to deformed beam axis
- bending moment is proportional to curvature of the beam: $M(x) = -EI w''$
- slender beams, where $L \gg B$

$$d) w'' = \frac{M(x)}{EI}$$



$$EI w'' = P$$

$$EI w' = Px + C_1$$

$$-M(x) = EI w'' = \frac{P}{2} x^2 + C_1 x + C_2$$

$$EI w' = \frac{P}{6} x^3 + \frac{C_1}{2} x^2 + C_2 x + C_3$$

$$EI w = \frac{P}{24} x^4 + \frac{C_1}{6} x^3 + \frac{C_2}{2} x^2 + C_3 x + C_4$$

$$w(0) = 0 \Rightarrow C_4 = 0$$

$$w'(0) = 0 \Rightarrow C_3 = 0$$

$$w(2a) = 0: 0 = \frac{P}{24} (2a)^4 + \frac{C_1}{6} (2a)^3 + \frac{C_2}{2} (2a)^2$$

$$M(2a) = 0: 0 = \frac{P}{2} (2a)^2 + C_1 2a + C_2$$

$$\Rightarrow 2Pa^3 + 2aC_1 + C_2 = 0 \quad \text{II}$$

$$\text{I: } \frac{2}{3} Pa^4 + \frac{4}{3} a^3 C_1 + 2 C_2 a^2 \cdot 0 / 2a^2$$

$$\frac{1}{3} Pa^4 + \frac{2}{3} a^3 C_1 + C_2 \cdot 0$$

$$\text{I II: } \frac{5}{3} Pa^3 + \frac{4}{3} a C_1 = 0$$

$$\Rightarrow C_1 = -\frac{5Pa^2}{3} = -\frac{15Pa^2}{12a} = -\frac{5}{4} Pa \cdot C_1$$

$$\Rightarrow C_2 = -2Pa^2 - 2a \left(-\frac{5}{4} Pa \right)$$

$$= -2Pa^2 + \frac{5}{2} Pa^2$$

$$\Rightarrow C_2 = \frac{1}{2} Pa^2$$

$$\Rightarrow M(x) = -\frac{P}{2} x^2 + \frac{5}{4} Pa x - \frac{1}{2} Pa^2$$

$$M(0) = -\frac{1}{2} Pa^2$$

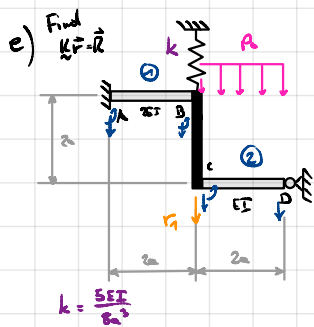
$$M(2a) = -2Pa^2 + \frac{5}{2} Pa^2 - \frac{1}{2} Pa^2 =$$

$$= 0 \quad \checkmark$$

Comparison

$$\text{from beam table: } M_A = -\frac{PL^2}{8} = -\frac{P(2a)^2}{8} = -\frac{Pa^2}{2} \quad \checkmark$$

COMPARISON



① $\underline{V} = \underline{Q} \underline{T}$

$$\underline{Q} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

② $\underline{Q} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

FORCE VEC

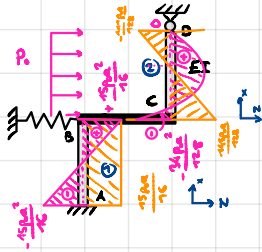
$\underline{R}^{(n)} = [0] \rightarrow$ no concentrated forces

② $\underline{S}^{(n)} = \frac{pL}{8} \begin{bmatrix} -5 \\ L \\ -3 \end{bmatrix}$ $\underline{R} = \underline{Q}^{(n)T} \underline{S}^{(n)} = [1 \ 0 \ 0] \frac{pL}{8} \begin{bmatrix} -5 \\ L \\ -3 \end{bmatrix} = \underline{-\frac{5}{8}pL} = \underline{\underline{R}^{(n)}}$

$\Rightarrow \underline{R}^{(n)} = \underline{R}^{(n)} = -\frac{5pL}{8}$
 $\Rightarrow \underline{R} = -\underline{R}^{(n)} = \underline{\frac{5pL}{8}} = \underline{\frac{5pa}{4}} = \underline{\underline{R}}$

System: $\underline{k} \underline{r} = \underline{R} \Rightarrow \frac{4EI}{a^3} r = \frac{5pa}{4} \Rightarrow r = \frac{5pa^4}{16EI}$

d) Find M, Q for AB & CD; maximo, slope & curv. for CD



① $\underline{S}^{(n)} = \underline{k}^{(n)} \underline{Q}^{(n)} r = \frac{2EI}{L^3} \begin{bmatrix} 12 & -6L & -12 & -6L \\ -6L & 4L^2 & 6L & 2L^2 \\ -12 & 6L & 12 & 6L \\ -6L & 2L^2 & 6L & 4L^2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} r$

$$= \frac{2EI}{L^3} \begin{bmatrix} -12 \\ 6L \\ 12 \\ 6L \end{bmatrix} r = \frac{2EI}{a^3} \begin{bmatrix} -12 \\ 6a \\ 12 \\ 6a \end{bmatrix} \frac{5pa^4}{16EI} = \frac{60pa}{64} \begin{bmatrix} -1 \\ a \\ 1 \\ a \end{bmatrix} = \frac{15pa}{16} \begin{bmatrix} -1 \\ a \\ 1 \\ a \end{bmatrix} = \underline{\underline{\underline{S}^{(n)}}}$$

V_A, M_A, V_B, M_B

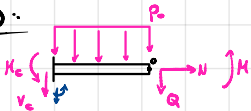
AB:



$\Sigma F_y = 0: Q - V_A = 0 \Rightarrow Q = V_A = \underline{\underline{+\frac{15pa}{16}}} = Q_{AB}$

$\Sigma M = 0: M(x) + K_A + V_A = 0 \Rightarrow M(x) = -K_A - V_A = \underline{\underline{-\frac{15pa^2}{16} + \frac{15pa}{16}x}}$

CD:



$\Sigma F_y = 0: V_C + Q + p \cdot x = 0 \Rightarrow Q = -V_C - p \cdot x = \underline{\underline{-\frac{145pa}{128} - p \cdot x}}$

$\Rightarrow Q_C = \underline{\underline{-\frac{145pa}{128}}}$

$\Rightarrow Q_D = \underline{\underline{-\frac{141pa}{128}}}$

$\Sigma M = 0: M(x) + K_C + V_C \cdot x + \frac{p \cdot x^2}{2} \Rightarrow M(x) = -K_C - V_C \cdot x - \frac{p \cdot x^2}{2} = \underline{\underline{-\frac{34a^2 p}{128} - \frac{34a^2 p}{128} - \frac{290pa^2}{128} = 0 \checkmark}}$

① $\underline{k}_{col} = \frac{2EI}{L^3} \begin{bmatrix} 12 & -6L & -12 & -6L \\ -6L & 4L^2 & 6L & 2L^2 \\ -12 & 6L & 12 & 6L \\ -6L & 2L^2 & 6L & 4L^2 \end{bmatrix}$

$\underline{k}^{(n)} = \underline{Q}^{(n)T} \underline{k}^{(n)} \underline{Q}^{(n)}$

$= [0 \ 0 \ 1 \ 0] \cdot \frac{2EI}{L^3} \begin{bmatrix} 12 & -6L & -12 & -6L \\ -6L & 4L^2 & 6L & 2L^2 \\ -12 & 6L & 12 & 6L \\ -6L & 2L^2 & 6L & 4L^2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$
 $= \frac{2EI}{L^3} [-12 \ 6L \ 12 \ 6L] \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \frac{2EI}{L^3} \cdot 12 = \underline{\underline{\frac{24EI}{L^3} = \frac{3EI}{a^3} = \underline{\underline{k^{(n)}}}}}$

② $\underline{k}_{beam} = \frac{3EI}{L^3} \begin{bmatrix} 1 & -L & -1 \\ -L & L^2 & L \\ -1 & L & 1 \end{bmatrix}$

$\underline{k}^{(n)} = [1 \ 0 \ 0] \frac{3EI}{L^3} \begin{bmatrix} 1 & -L & -1 \\ -L & L^2 & L \\ -1 & L & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \frac{3EI}{L^3} [1 \ -L \ 1] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\Rightarrow \underline{\underline{k^{(n)}}} = \underline{\underline{\frac{3EI}{a^3}}}$

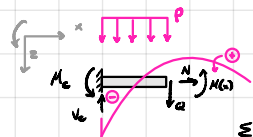
GLOBAL: $\underline{K} = \underline{k}^{(n)} + \underline{k}^{(n)} + \underline{k} = \frac{3EI}{a^3} + \frac{3EI}{a^3} + \frac{5EI}{a^3} = \underline{\underline{\frac{32EI}{a^3} = \frac{4EI}{a^3} \cdot K}}$

CD:

• As $Q = \frac{dM}{dx}$, it is expected that M has a maximum where Q is 0.

• The slope of Q is $-p$, curvature is zero

• The slope of M is Q , the curvature is $-p$



$$M_c = \frac{PL^2}{8} = \frac{Pa^2}{2}$$

$$V_c = \frac{5Pa}{8}$$

$$= \frac{5Pa}{4}$$

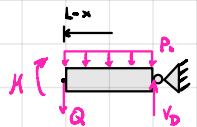
$$\sum M = 0: M(x) + M_c - V_c \cdot x + \frac{P \cdot x^2}{2} = 0$$

$$: M(x) + \frac{Pa^2}{2} - \frac{5Pa}{4}x + \frac{Px^2}{2} = 0$$

$$\Rightarrow M(x) = -\frac{Px^2}{2} + \frac{5Pa}{4}x - \frac{Pa^2}{2}$$

$$M_c = M(0) = -\frac{Pa^2}{2}$$

$$M_D = M(2a) = 0$$



$$V_D = \frac{3PL}{8} = \frac{3Pa}{4}$$

$$\sum M = 0: M(x) - \frac{P(L-x)^2}{2} + V_D(L-x) = 0$$

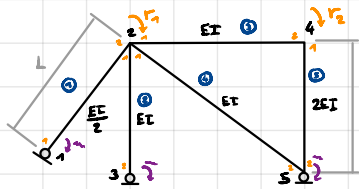
$$M(x) = \frac{P(L-x)^2}{2} + \frac{3Pa}{4}(L-x)$$

$$M(x=2a) = 0$$

$$M(x=0): -\frac{4Pa^2}{2} + \frac{6Pa^2}{4} = -\frac{Pa^2}{2} = M_c$$

TEST ONLY

Problem 2)



a)

$$\text{nodal (e) dof: } [\theta_1, x_2, z_2, \theta_3, \theta_3, x_4, z_4, \theta_4, \theta_5]^T$$

$$\text{global dof: } r = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5]^T$$

difference between 1,3,5 & 2,4: 1,3,5 have their translational dof's restricted by boundary conditions,
2,4 ————— " ————— by not considering axial deformations.
all of them can only rotate for that reason.

b) Find $\tilde{K}$

$$\tilde{S}^{(e)} = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & 12 & 6L \\ -6L & 4L^2 & 6L & 2L^2 \\ 12 & 6L & 12 & 6L \\ -6L & 2L^2 & 6L & 4L^2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$$

$$\Rightarrow \tilde{k}^{(e)} = \frac{2EI}{L} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

	1	2	3	4	5
1	1	2	2	2	4
2	2	3	4	5	5

$$k_1 = \frac{EI}{L}$$

$$k_2 = \frac{2EI}{L}$$

$$k_3 = \frac{2EI}{L_{24}}$$

$$k_4 = \frac{2EI}{L_{25}}$$

$$k_5 = \frac{4EI}{L}$$

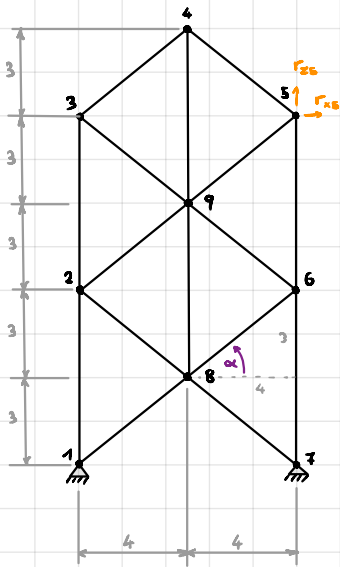
$$\Rightarrow \tilde{K} = \begin{bmatrix} 2k_1 & k_1 & 0 & 0 & 0 \\ k_1 & 2k_1+2k_3+2k_4 & k_2 & k_3 & k_4 \\ 0 & k_2 & 2k_2 & 0 & 0 \\ 0 & k_3 & 0 & 2k_3+2k_5 & k_5 \\ 0 & k_4 & 0 & k_5 & 2k_4+2k_5 \end{bmatrix}$$

c) How does $\tilde{K}$ change if θ_2 is restrained?

Row and column 2 become 0, element K_{22} becomes 1.

K_{22} also becomes 0.

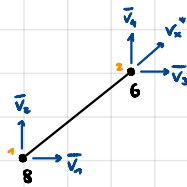
Problem 3:



$$EA = 50\,000 \text{ kN}$$

$$\vec{r}_i = [r_{xi} \ r_{zi}]^T; \quad i = 1, \dots, 9$$

Element 8-6



$$a) \text{ find } \underline{T}_{8-6}: \vec{v}_{e,e} = v_{ei} \cdot [T_1 \ T_2 \ T_3 \ T_4] \begin{bmatrix} \vec{v}_1 \\ \vec{v}_2 \\ \vec{v}_3 \\ \vec{v}_4 \end{bmatrix} = \underline{T}_{8-6} \vec{\tilde{v}}_{e,e}$$

$$\underline{T}_0 = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix}, \quad \alpha = \tan^{-1}\left(\frac{3}{4}\right) =$$

$$\Rightarrow \underline{T}_8 = \begin{bmatrix} \underline{T}_0 & \underline{0} \\ \underline{0} & \underline{I}_0 \end{bmatrix}$$